AI Engineer
GenAI & LLM Solutions

DevOps
PhD in Mathematics

Francisco Cristóvão Agostinho

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AI Engineer with a PhD in Mathematics, building production LLM solutions for the banking industry — financial analysis of companies, intelligent document processing and fraud analytics. I design and ship end-to-end GenAI systems on Google Cloud: RAG pipelines, OCR and document understanding, packaged as internal Python libraries so that other teams can reuse them safely.

Some experience with CI/CD with GitHub Actions, reusable team templates, containerised runners and automated application deployments, making sure solutions are deployed safely and upholding to compliance regulations.

My mathematical background in statistics, probability theory and differential equations underpins how I evaluate and reason about model behaviour, while years of teaching mean I can explain complex AI systems clearly to engineers, business stakeholders and risk/compliance audiences alike.

Skills

AI / GenAI — LLM application design, RAG, prompt engineering, LLM evaluation & guardrails, embeddings & vector search, document AI / OCR (Docling, Document AI, Tesseract, EasyOCR)

Cloud (Google Cloud only) — Vertex AI, Gemini, Cloud Run, Cloud Storage, BigQuery, Document AI, Artifact Registry, Secret Manager, IAM & Workload Identity Federation, Cloud Logging

Engineering — Python (FastAPI, Pydantic, pytest, uv), SQL / MySQL, REST APIs, MCP servers, Docker, Git, GitHub Actions, CI/CD pipelines, internal library packaging & release management

Data & ML — NumPy, Pandas, scikit-learn, PySpark, Matplotlib, Seaborn, statistics & probability theory, anomaly detection

Professional Experience

- **Caixa Geral de Depósitos — Portugal**
  AI Engineer: 2026 – currently
  - Design and deliver LLM-based solutions for banking, covering financial analysis of companies, document processing and fraud analysis.
  - Build RAG and document-understanding pipelines that turn unstructured banking documents into structured, queryable data.
  - Develop agentic workflows and evaluation harnesses so LLM outputs are reliable, measurable and auditable in a regulated environment.
  - Author and maintain shared internal Python libraries (LLM clients, RAG, OCR, monitoring, database adapters) published to GCP Artifact Registry and consumed across teams.
  - Hands on in DevOps / CI-CD layer of our team: GitHub Actions workflows, reusable repository templates, custom runner images and automated deployments to Cloud Run, Cloud Jobs and Cloud Tasks.
  - Work exclusively on Google Cloud — Vertex AI and Gemini for models, Cloud Run for services, Artifact Registry for packaging, GCP-native IAM, secrets and observability.

  Attained goals:
  - Developed a GenAI solution for financial analysis of enterprises - a project evaluated as having at least 1 million euros of value to the company and being actively used by the entire risk department.
- **Nova School of Business and Economics (NOVA) — Portugal**
  Teaching Assistant: 2023 – 2026
  - Linear Algebra with Programming and Calculus courses for Management and Economics students.
  - Teamwork in teams of 8 to 10 people to devise and execute challenging and stimulating lessons, constructing and grading exams, reporting feedback to students.
  - Deepened my communication skills, both peer related and student related.

  Attained goals:
  - 94% student satisfaction — average of 5.64 out of 6, above the school average of 5.18.
- **Faculty of Science of the University of Lisbon — Portugal**

Teaching Assistant: 2018-2020
- Linear Algebra and Calculus for B.Sc. in biochemistry, statistics and engineering.

Education

- **Instituto Superior Técnico — Portugal**
  - PhD in Mathematics. Final dissertation: *Classification and stability results of ground state solutions on metric graphs* – 2020-2024
- **Faculty of Science of the University of Lisbon — Portugal**
  - M.Sc. in Mathematics. Final dissertation: *Existence and qualitative properties of solutions to nonlinear Schrödinger equations on metric graphs* – 2017-2020
    - Peer elected student body delegate
    - Applied and obtained a research scholarship
  - B.Sc. in Mathematics – 2013-2017
    - Peer elected student body delegate

Selection of Publications and Talks

- **Publication:** Classification and stability of positive solutions to the NLS equation on the T-metric graph

Agostinho, F., Correia, S., Tavares, H.
Nonlinearity, 2024

- **Publication:** A comprehensive studyof bound states for the nonlinear Schrödinger equation on single-knot metric graphs

Agostinho, F., Correia, S., Tavares, H.
Proceedings of the London Mathematical Society, 2025

- **Publication:** Stability transitions of NLS action ground-states on metric graphs

Agostinho, F., Correia, S., Tavares, H

- **International Talk:** NLS positive bound states on the T -graph

Encontro Conjunto Brasil-Portugal em Matemática
Universidade Federal da Bahia, Salvador da Bahia, Brazil

Languages

Portuguese (Native) | **English** (Fluent) | **Dutch** (Basic) | **Japanese** (Basic)